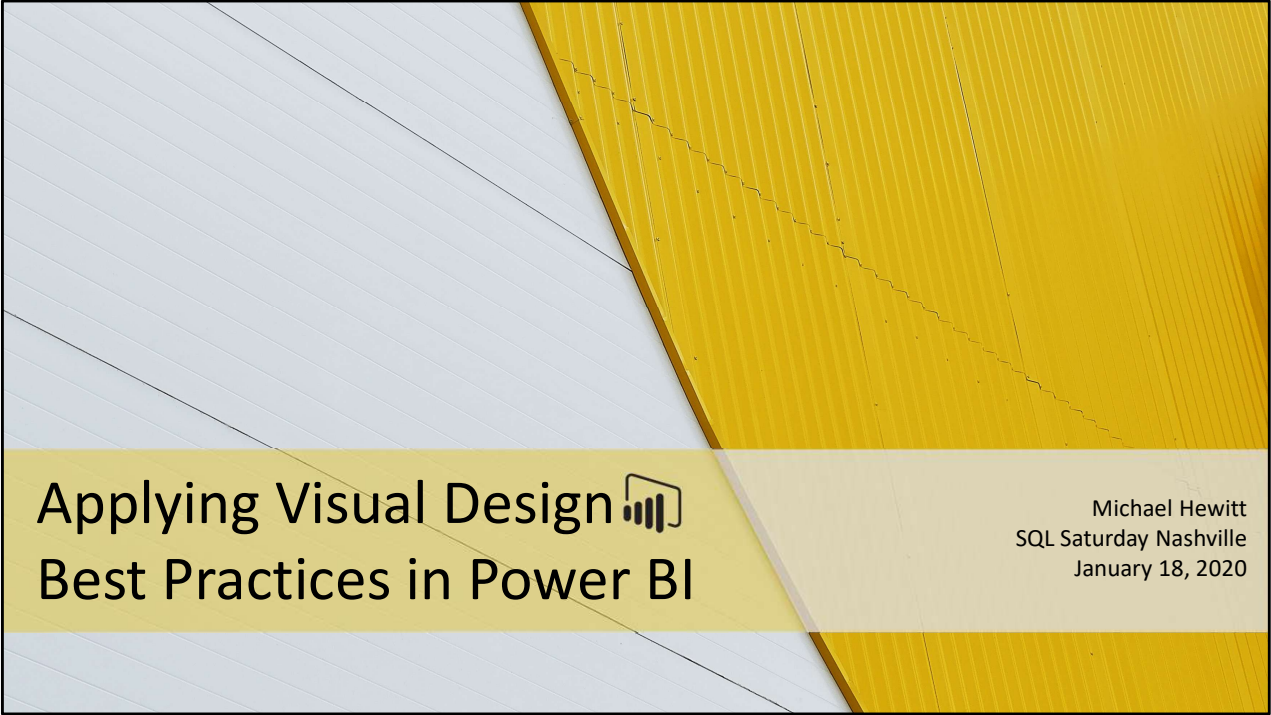




Applying Visual Design Best Practices in Power BI

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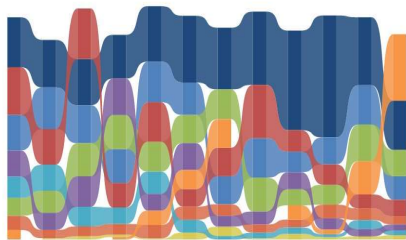


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Total Time by Month and Category





How many Power BI users? Other visualization tools?
Analysts? Developers? DBAs? Report writers?

AGENDA

Report Design and Layout

Designing and Visualizing for Different Audiences

Implementing Custom Report Themes

There are many best practices in report design, these 3 offer a good starting point to springboard from

REQUIREMENTS

1. Traffic accident trend during the last decade
2. Top 10 accident locations
3. See count of accidents by lighting condition (dark, dusk, dawn, day)
4. Filter by Hit & Run (Y/N/Unk), Injury Accident (Y/N/Unk), Fatalities (Y/N/Unk), Weather Condition, & Number of Motor Vehicles Involved

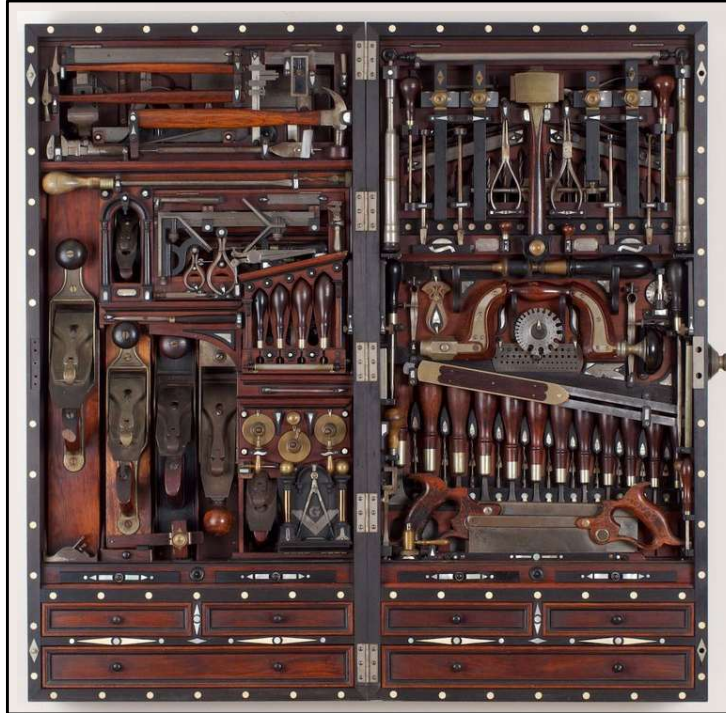
Gathering good requirements is necessary to a good design. Meet with key stakeholders to understand their pain points and needs for reporting. Interview SME's to gain understanding of business processes, the data, and calculations. Create a storyboard and wireframe and present to the stakeholders prior to development.

Metro Nashville traffic accidents data from [Data.Nashville.gov](https://data.nashville.gov)

What is the best way to present this data
so that the user can understand the story
and take action?

There is a lot of research being done in data visualization, how do we take that great work that is being done and apply that in our organization?

As report writers/designers, often times end users are looking to us to help them find the insights in their data



REPORT DESIGN and LAYOUT

Photo: Henry Studley Tool Chest

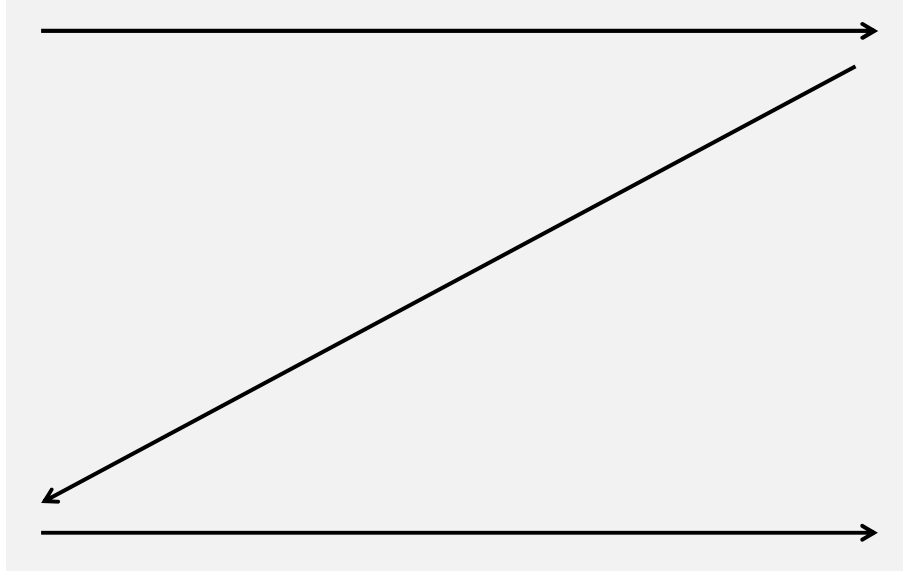
Attention to detail in laying out a report well goes a long way in user experience



Imagine that this slide is a blank desktop or tablet report canvas

Title – Every page should have a descriptive or action title. Give users a context of the data that is on being displayed and indicate how the user should take action.

Title



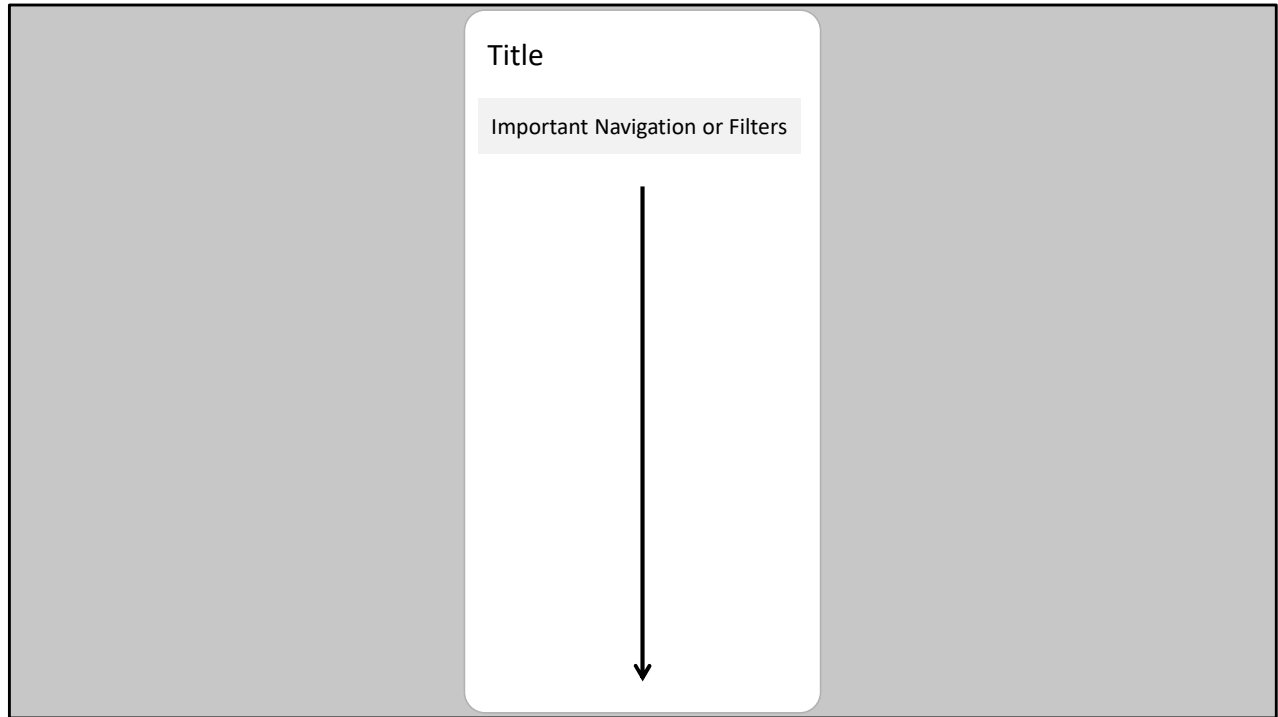
Move from left to right, top to bottom across the screen

- The most important visuals at the top left, least important at the bottom right



Navigation and filter elements can live at the top, left, or right side of the screen depending on your requirements or preferences.

- Will the user need to make selections in a slicer to draw meaning from the visuals on the page?
- Are there navigation elements, like a back button on a drill through page or buttons that control bookmarks, that are a critical part of the UX?



Mobile Design

- In a portrait layout, generally top to bottom
- Power BI's mobile reports work but the experience is a bit clunky – report design is not as flexible
- The key is to keep it simple
- Don't try to force all of the content from a desktop report into a phone screen. Choose the data/visuals that are most important or that a user would need to monitor on the go and include those.
- 1 or 2 important filters, more than that make the report feel bulky
- Use responsive mode on the visuals

DEMO

Layout report using these best practices
Create mobile report to demonstrate pros & cons



Designing for different audiences

Conditional formatting on tables – be careful of clutter!

Add visuals along with a table to say the same thing – often the more “visual” visuals may not be the preference of your audience but have a great impact and can help to increase the visual literacy in the organization. Use bookmarks to toggle between the graph and table.

Photo by John Schnobrich

DEMO

Add table and create buttons to toggle back and forth between visuals – Demonstrate “wiring up” the bookmarks



Photo by Marko Blazevic

DEMO

Color.adobe.com

Power BI Desktop December 2019 release – turn on the preview feature!

<https://powerbi.tips/tools/report-theme-generator-v3/>



Photo by Artem Maltsev

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3



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